A Seeded Sensitivity Study of Utilization and Locality in a Simplified Page-Mapped Flash Simulator

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Abstract

We report a reproducible sensitivity study of a small page-mapped flash translation layer simulator. Across 30 seeded trials, we varied active logical-page utilization (40%, 70%, 85%) and update locality (uniform or an 80% hot-set-biased mixture). Mean amplification rose from 1.132 to 4.106 for uniform workloads as utilization increased from 40% to 85%. Hot/cold means were 1.338, 1.821, and 2.172. This interaction is specific to the greedy allocator and synthetic workload. This is a software sensitivity result, not a novel FTL algorithm, hardware measurement, or SSD lifetime prediction. A previously reported record-buffer comparison is withdrawn because its paths did not preserve equivalent logical state.

1. Introduction

Flash translation layers map host logical addresses to physical NAND pages and reclaim invalid pages through garbage collection. Page relocation makes write amplification dependent on workload and device policy. Prior work analyzes these effects in SSDs [1, 2]. We ask how active logical-page utilization and update locality affect modeled program amplification in an educational model.

2. Related Work

Hu et al. analyze write amplification in flash-based SSDs [1]. Agrawal et al. describe architectural tradeoffs in flash SSDs [2]. This establishes that the general topic is not novel. Our contribution is a limited reproducible sensitivity artifact with corrected measurement boundaries.

3. Simulation Architecture

Each physical page is free, valid, or invalid; logical-to-physical mappings reside in an in-memory dictionary. Writes program a free page and invalidate a prior page for that LBA. Greedy GC selects a block with many invalid pages, copies valid pages, and erases it, breaking ties by lower erase count. Persistent mapping metadata, NAND timing, parallelism, bad blocks, ECC, failures, and static wear leveling are omitted. The supplied ring buffer discards record LBAs and cannot be compared as an equivalent key-value store.

4. Methodology

We used 16 blocks × 32 pages, 4,096-byte pages, and 5,000 full-page host overwrites per trial. Active address spaces were 40%, 70%, and 85% of physical pages. Uniform trials sample active LBAs uniformly. Hot/cold trials direct 80% of updates to 10% of the active range and 20% uniformly over the full range; these components can overlap. Five paired seeds (11, 23, 47, 89, 101) yield 30 trials. We report A=(H+G)/H, where H is host page programs and G is GC copy programs, in modeled page programs per host page program. Results combine startup and GC; no steady-state convergence is claimed. SD is across five seeds.

5. Results

Uniform amplification increased with utilization. Hot-set-biased amplification was higher at 40%, but lower at 70% and 85% than uniform traffic. This interaction is specific to the allocator, victim score, and address distribution.

Table 1. Program amplification, mean ± sample SD across five seeds.

Utilization	Uniform	Hot/cold
40%	1.132 ± 0.002	1.338 ± 0.016
70%	1.988 ± 0.013	1.821 ± 0.073
85%	4.106 ± 0.067	2.172 ± 0.179

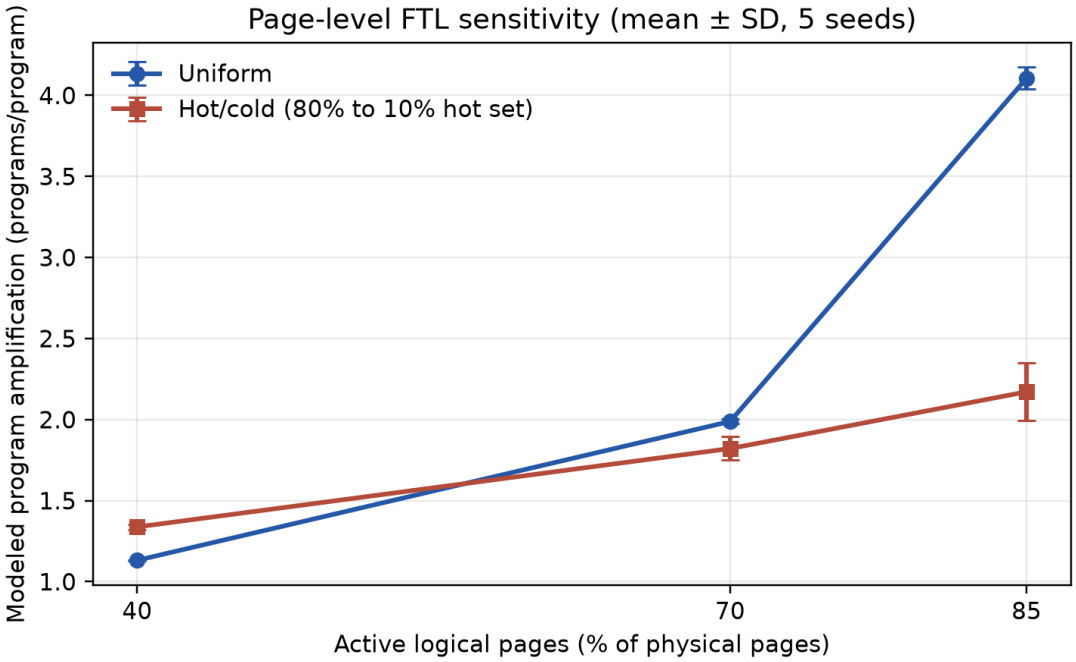


Figure 1. Modeled program amplification by active logical-page utilization; bars show one sample SD across five seeds.

6. Discussion

The increase for uniform traffic at high utilization is consistent with less spare capacity making reclamation more expensive in this model. The differing hot/cold trend demonstrates interaction with greedy victim selection and allocation order. Small across-seed SD does not establish robustness to other workloads or FTL algorithms. This is not device WAF validated against hardware.

7. Limitations

This abstraction uses an in-memory mapping table and omits persistent recovery, controller caches, metadata writes, wear-leveling migration, timing, parallel channels, bad blocks, endurance failures, and hardware validation. Only six parameter cells, five seeds, and one run length were studied. Erase counts are not lifetime estimates. The 66-fold buffer comparison is withdrawn; crash outputs are queue occupancy accounting, not recovery results.

8. Conclusion

In this simplified page-mapped simulator, utilization and locality jointly affected modeled GC amplification, with locality ordering changing across utilization. This is an educational sensitivity study, not a novel FTL technique or commercial flash prediction.

Code and Data Availability

Source, tests, configuration, raw trials, and figure scripts: https://github.com/apandya55/flash-waf-sim. No archival DOI exists. Reproduction instructions are in research/METHODOLOGY.md.

AI Use Disclosure

AI tools assisted with auditing, code editing, experiment scripting, and manuscript drafting. The author must verify all content and adapt disclosure to the target journal.

References

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